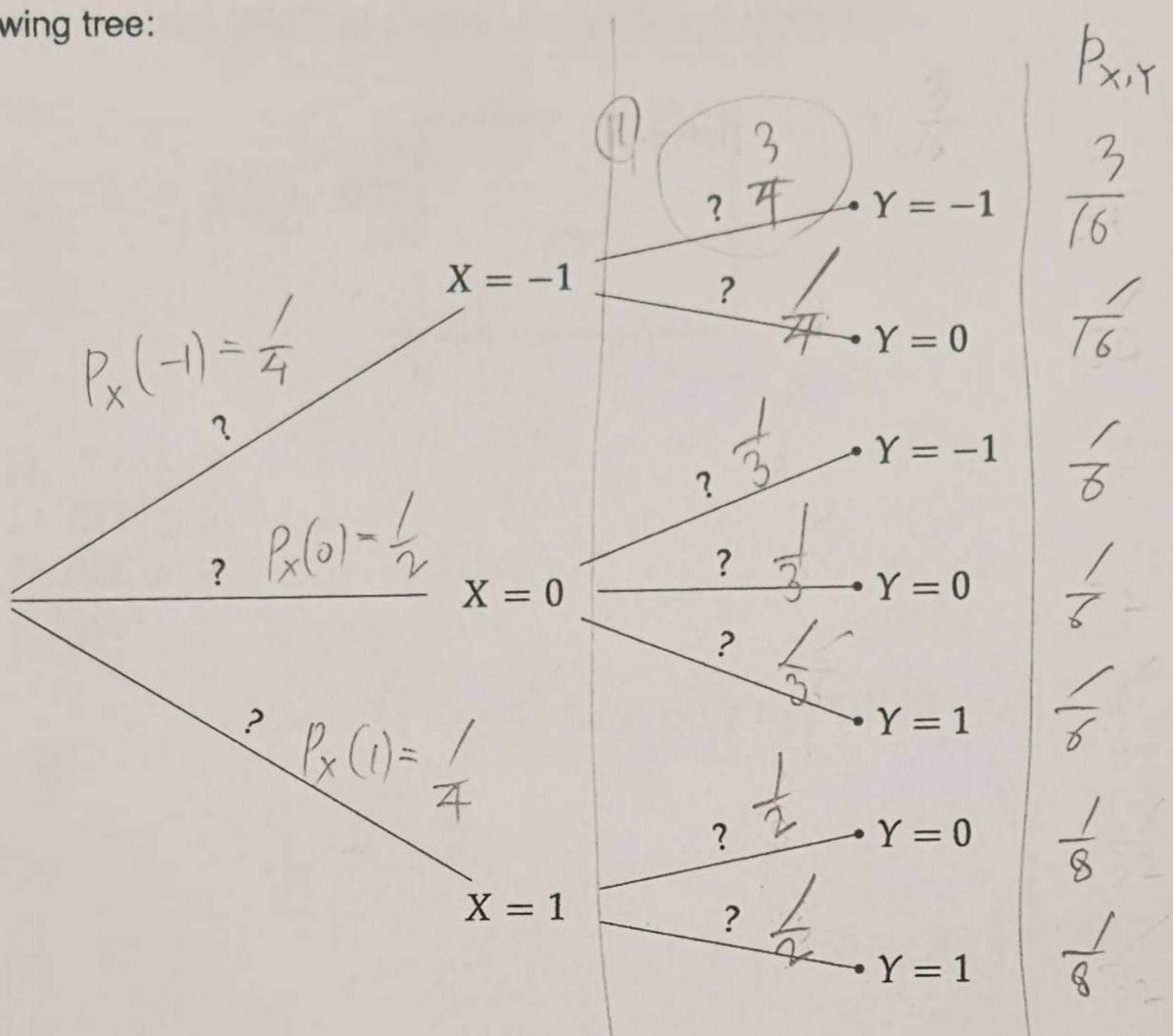
b. The experiment from which X and Y are derived is performed sequentially. First, X is found, then Y is found. In this context, label the conditional branch probabilities of the following tree:



① $P(Y=-1 \mid X=-1) = \dfrac{P(Y=-1, X=-1)}{P(X=-1)} = \dfrac{P_{X,Y}(-1,-1)}{P_X(-1)} = \dfrac{\frac{3}{16}}{\frac{1}{4}} = \dfrac{3}{4}$

③ $P(Y=0 \mid X=-1) = \dfrac{\frac{1}{16}}{\frac{1}{4}} = \dfrac{1}{4}$

② $P(Y=-1 \mid X=0) = \dfrac{\frac{1}{6}}{\frac{1}{2}} = \dfrac{1}{3}$

$P(Y=0 \mid X=0) = \dfrac{\frac{1}{6}}{\frac{1}{2}} = \dfrac{1}{3}$

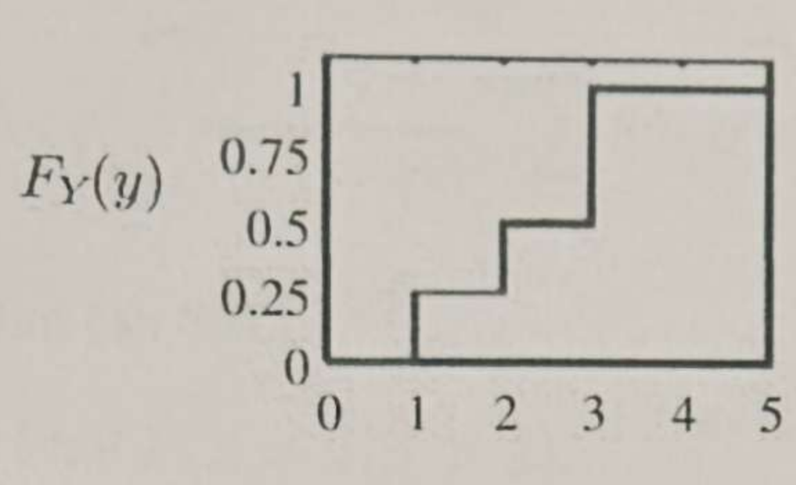
$P(Y=1 \mid X=0) = \dfrac{\frac{1}{6}}{\frac{1}{2}} = \dfrac{1}{3}$

$P(Y=0 \mid X=1) = \dfrac{\frac{1}{8}}{\frac{1}{4}} = \dfrac{1}{2}$

$P(Y=1 \mid X=1) = \dfrac{\frac{1}{8}}{\frac{1}{4}} = \dfrac{1}{2}$

5. (10 points) Let Y be a discrete random variable whose cumulative distribution function (CDF) shown in the figure below. Compute the following probabilities, and describe the probability mass function (PMF) of Y either analytically or graphically.

10.



a. $P(Y < 1), P(Y \leq 1)$
b. $P(Y > 2), P(Y \geq 2)$
c. $P(Y = 3), P(Y > 3)$
d. PMF of Y, $P_Y(y)$

$P(Y=1) = 0.25$

$P(Y=2) = 0.25$

$P(Y=3) = 0.5$

$P(\text{else}) = 0$

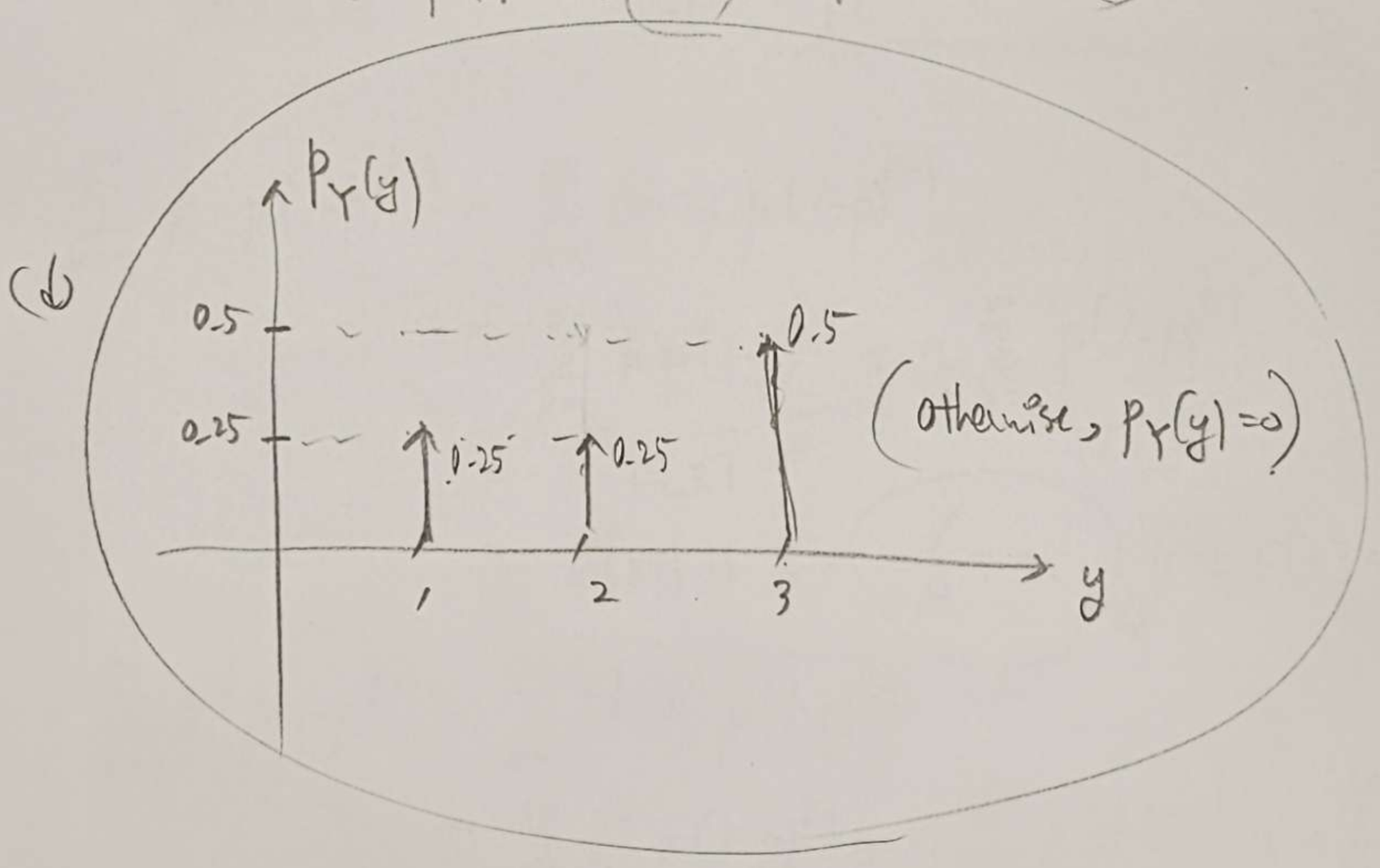
(a) $P(Y<1) = 0$ $P(Y\leq 1) = P(Y=1) = 0.25$

(b) $P(Y>2) = P(Y=3) = 0.5$

$P(Y\geq 2) = P(Y=2) + P(Y=3) = 0.25+0.5 = 0.75$

(c) $P(Y=3) = 0.5$ $P(Y>3) = 0$

(d)



1-p = q

q p = 1-q

6. (16 points) Let X be a geometric random variable with parameter p, where $0 < p < 1$, and $P(X = k) = (1-p)^{k-1}p$, $k = 1,2,3,\ldots$. Answer the following:

a. Starting from the identity $\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$, $|q| < 1$, derive the following results by differentiation: $\sum_{k=1}^{\infty} k \cdot q^{k-1} = \frac{1}{(1-q)^2}$, $\sum_{k=1}^{\infty} k^2 \cdot q^{k-1} = \frac{1+q}{(1-q)^3}$.

b. Using the result in part (a), find $\mathbb{E}[X^2]$.

c. Find $\mathbf{Var}(X)$.

d. Find the conditional PMF $P(X = k \mid X > 2)$.

e. Find $\mathbb{E}[X \mid X > 2]$.

f. Find $\mathbf{Var}[X \mid X > 2]$.

(b) $E[X^2] = \sum_{k=1}^{\infty} k^2 \cdot p(1-p)^{k-1}$ 에서 $p = 1-q$ 라는 정보

$E[X^2] = \sum_{k=1}^{\infty} k^2 \cdot (1-q) \cdot q^{k-1}$

$= (1-q) \underbrace{\sum_{k=1}^{\infty} k^2 \cdot q^{k-1}}_{(\text{식 4})} = (1-q) \cdot \frac{1+q}{(1-q)^3}$

$= \frac{1+q}{(1-q)^2}$

$= \frac{2-p}{p^2}$ ☆

(c) $\mathrm{Var}[X] = E[X^2] - (E[X])^2 = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}$

(d) $X > 2$ 라는 이제 $(1-p)$ 동전이 2번 성공 → $P(X > 2) = (1-p)^2$

$P(X = k \mid X > 2) = \frac{P(X=k)}{P(X>2)} = \frac{(1-p)^{k-1} \cdot p}{(1-p)^2} = p(1-p)^{k-3} \quad (k \geq 3)$ ☆

(e) $E[X \mid X > 2] = \sum_{k=3}^{\infty} k \cdot p(1-p)^{k-3} = \sum_{k=1}^{\infty} (k+2)\, p(1-p)^{k-1}$

$= \underbrace{\sum_{k=1}^{\infty} k p (1-p)^{k-1}}_{E[X]} + 2 \underbrace{\sum_{k=1}^{\infty} p(1-p)^{k-1}}_{(=1)}$

$= E[X] + 2 = \frac{1}{p} + 2$ ☆ $(\because E[X] = \frac{1}{p})$

(f) $E[X^2 \mid X > 2] = \sum_{k=3}^{\infty} k^2 p(1-p)^{k-3} = \sum_{k=1}^{\infty} (k+2)^2 p(1-p)^{k-1}$

$= \underbrace{\sum_{k=1}^{\infty} k^2 p(1-p)^{k-1}}_{\frac{1-p}{p^2}} + 4 \underbrace{\sum_{k=1}^{\infty} k p(1-p)^{k-1}}_{\frac{1}{p}} + \underbrace{4 \sum_{k=1}^{\infty} p(1-p)^{k-1}}_{4}$

$= E[X^2] + 4E[X] + 4$

$= \frac{2-p}{p^2} + \frac{4}{p} + 4 = \frac{2-p+4p+4p^2}{p^2}$

$= \frac{4p^2+3p+2}{p^2}$

$\mathrm{Var}[X \mid X > 2]$

$= E[X^2 \mid X > 2] - (E[X \mid X > 2])^2$

$= \frac{4p^2+3p+2}{p^2} - \left(\frac{1}{p}+2\right)^2$

$= \frac{4p^2+3p+2}{p^2} - \frac{1}{p^2}(4p^2+4p+1) = \frac{4p^2+3p+2-4p^2-4p-1}{p^2} = \frac{1-p}{p^2}$ ☆

7. (14 points) Let X and Y be discrete random variables with the following joint PMF:

$$p_{X,Y}(x,y) = \begin{array}{c|ccc} & y=0 & y=1 & y=2 \\ \hline x=0 & \frac{1}{8} & \frac{1}{8} & 0 \\ x=1 & \frac{1}{8} & \frac{1}{4} & \frac{1}{8} \\ x=2 & 0 & \frac{1}{8} & \frac{1}{8} \end{array}$$

and $p_{X,Y}(x,y) = 0$ otherwise. Answer the following:

a. Find the marginal PMFs $p_X(x)$ and $p_Y(y)$.

b. Determine whether X and Y are independent.

c. Let $A = \{(X,Y) : X+Y=2\}$. Find the conditional joint PMF $p_{X,Y|A}(x,y)$, and the conditional marginal PMFs $p_{X|A}(x)$ and $p_{Y|A}(y)$.

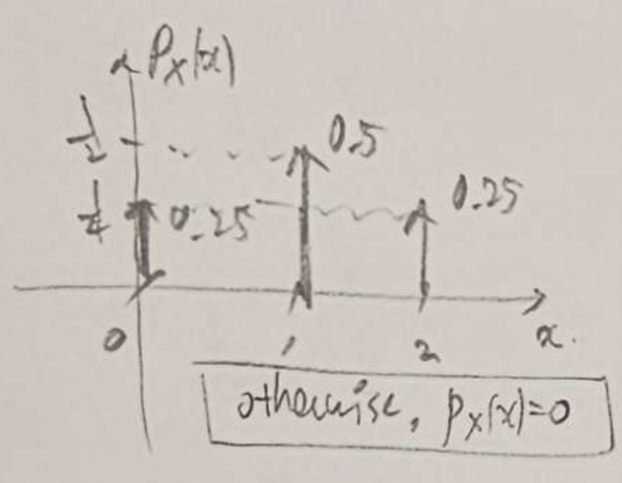
d. Show that X and Y are conditionally independent given A.

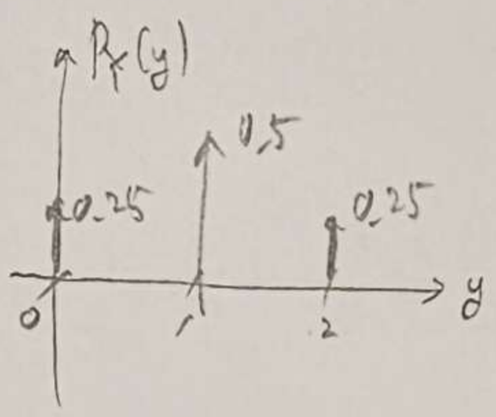
e. Compute $\mathbb{E}[X]$ and $\mathbb{E}[Y]$.

f. Compute $\mathbb{E}[X|A]$ and $\mathbf{Var}[Z|A]$.

g. Let $Z = 2X+3Y$. Compute $\mathbb{E}[Z|A]$ and

(a)





(g)

$E[2X+3Y|A]$

$= 2E[X|A] + 3E[Y|A]$

$= 2+3 = 5$

$E[(2X+3Y)^2|A] = 4E[X^2_{|A}] + 12E[XY_{|A}] + 9E[Y^2_{|A}]$

$= 4+12+9 = 25$ (16)

$\rightarrow \mathrm{Var}[Z|A] = 0$

(b) $P_{X,Y}(0,0) = \frac{1}{8}$ $P_X(0) = \frac{1}{4}$ $P_Y(0) = \frac{1}{4}$ $\rightarrow P_X(0)P_Y(0) = \frac{1}{16}$ $\rightarrow$ 독립 X

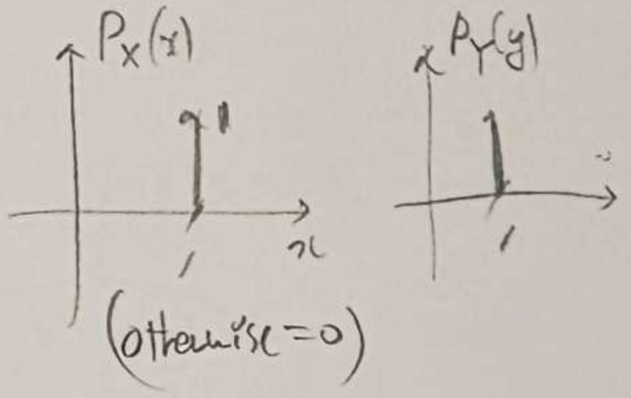
(c) $A = \{(0,2), (1,1), (2,0)\}$ $P(A) = P_{X,Y}(0,2) + P_{X,Y}(1,1) + P_{X,Y}(2,0) = 0 + \frac{1}{4} + 0 = \frac{1}{4}$

x \ y	0	1	2
0	-	-	0
1	-	1	-
2	0	-	-

$\rightarrow P_{X,Y|A}(0,2) = \frac{P_{X,Y}(0,2)}{P(A)} = 0$

$P_{X,Y|A}(1,1) = \frac{P_{X,Y}(1,1)}{P(A)} = 1$

$P_{X,Y|A}(2,0) = \frac{P_{X,Y}(2,0)}{P(A)} = 0.$



(d) $P_{X,Y|A}(0,2) = 0$ $P_{X|A}(0) = 0$ $P_{Y|A}(2) = 0$ $\rightarrow$ $P_{X,Y|A}(0,2) = P_{X|A}(0) \cdot P_{Y|A}(0)$

$P_{X,Y|A}(1,1) = 1$ $P_{X|A}(1) = 1$ $P_{Y|A}(1) = 1$ $\rightarrow$ $P_{X,Y|A}(1,1) = P_{X|A}(1) \cdot P_{Y|A}(1)$

$P_{X,Y|A}(2,0) = 0$ $P_{X|A}(2) = 0$ $P_{Y|A}(0) = 0$ $\rightarrow$ $P_{X,Y|A}(2,0) = P_{X|A}(2) \cdot P_{Y|A}(0)$

(e) $E[X] = 0 \times P_X(0) + 1 \times P_X(1) + 2 \times P_X(2)$

$= 0 \times \frac{1}{4} + 1 \times \frac{1}{2} + 2 \times \frac{1}{4} = \frac{1}{2} + \frac{1}{2} = 1$

$E[Y] = 0 \times P_Y(0) + 1 \times P_Y(1) + 2 \times P_Y(2)$

$= 0 + 1 \times \frac{1}{2} + 2 \times \frac{1}{4} = \frac{1}{2} + \frac{1}{2} = 1$

(f) $E[X|A] = 1 \times 1 = 1$

$E[X^2|A] = 1^2 \times 1 = 1$

$\mathrm{var}[X|A] = 1 - 1^2 = 0$

8. (12 points) A random variable X has the following PDF:

$$f_X(x) = \begin{cases} \frac{x}{2}, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

11

Answer the following:

a. Verify that $f_X(x)$ is a valid PDF.
b. Find the CDF $F_X(x)$.
c. Find $\mathbb{E}[X]$.
d. Find $\mathbb{E}[X^2]$.
e. Find $\mathbb{E}[e^X]$.
f. Find $P(X > 1)$.

(a) $\int_{-\infty}^{\infty} f_x(x)\,dx = \int_0^2 \frac{1}{2}x\,dx = \frac{1}{4}x^2\Big|_0^2 = \frac{1}{4}(4-0) = 1$ (valid)

(b) $F_x(x) = \int_{-\infty}^{x} f_x(t)\,dt = \int_0^x \frac{1}{2}t\,dt = \frac{1}{4}x^2$ $(0<x<2)$ → $F_x(x) \begin{cases} \frac{1}{4}x^2 & (0<x<2) \\ 0 & (x \le 0) \\ 1 & (x \ge 2) \end{cases}$

(c) $E[X] = \int_{-\infty}^{\infty} x f_x(x)\,dx = \int_0^2 x \cdot \frac{1}{2}x\,dx = \frac{1}{6}x^3\Big|_0^2 = \frac{1}{6}\cdot 8 = \frac{4}{3}$

(d) $E[X^2] = \int_{-\infty}^{\infty} x^2 f_x(x)\,dx = \int_0^2 x^2 \cdot \frac{1}{2}x\,dx = \frac{1}{8}x^4\Big|_0^2 = \frac{1}{8}\cdot 16 = 2$

(e) $E[e^X] = \int_{-\infty}^{\infty} e^x \cdot f_x(x)\,dx = \int_0^2 e^x \cdot \frac{1}{2}x\,dx$

$= \frac{1}{2}x\cdot e^x\Big|_0^2 - \int_0^2 \frac{1}{2}e^x\,dx$

$= \frac{1}{2}\cdot 2e^2 - \frac{1}{2}e^x\Big|_0^2 = e^2 - \frac{1}{2}(e^2-1)$

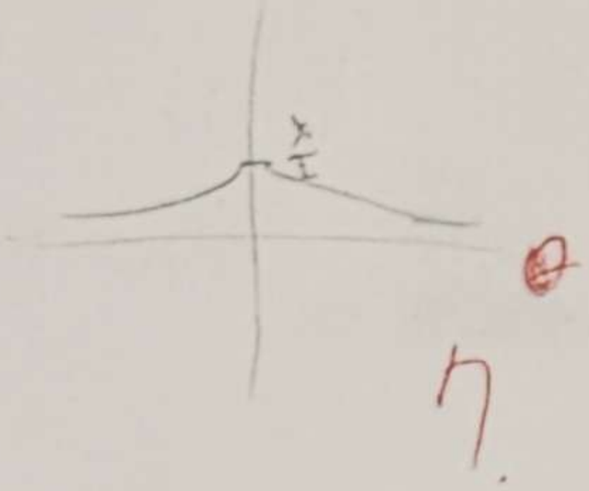
$= \frac{1}{2}e^2 + \frac{1}{2}$

(f) $P(X>1) = 1 - P(X \le 1)$

$= 1 - F_x(1) = 1 - \frac{1}{4} = \frac{3}{4}$

9. (12 points) Let Y be a continuous random variable with PDF

$$f_Y(y) = \frac{\lambda}{2} e^{-\lambda|y|}, \qquad -\infty < y < \infty,$$



where $\lambda > 0$. Answer the following:

a. Show that $f_Y(y)$ is a valid PDF.

b. Show that the distribution of Y is symmetric about 0.

c. Find $\mathbb{E}[Y]$.

d. Find $\mathbf{Var}[Y]$.

e. Find $P(|Y| > a)$ for $a \geq 0$.

f. For a real constant t, compute $\mathbb{E}[e^{tY}]$ for those values of t for which the expectation exists.

(a) $\int_{-\infty}^{\infty} f_Y(y)\,dy = \int_0^{\infty} \frac{\lambda}{2} e^{-\lambda y}\,dy + \int_{-\infty}^{0} \frac{1}{2} e^{\lambda y}\,dy$

$= -\frac{1}{2} e^{-\lambda y}\Big|_0^{\infty} + \frac{1}{2} e^{\lambda y}\Big|_{-\infty}^{0} = -\frac{1}{2}(0-1) + \frac{1}{2}(1-0) = 1$ (valid) ★

(b) $f_Y(y)$가 y축에 대하여 대칭이므로 $P(Y>0) = P(Y<0) \rightarrow$ 0 기준 대칭.

(c) $E[Y] = \int_0^{\infty} y \cdot \frac{\lambda}{2} e^{-\lambda y}\,dy + \int_{-\infty}^{0} y \cdot \frac{\lambda}{2} e^{\lambda y}\,dy$

$= -y \cdot \frac{1}{2} e^{-\lambda y}\Big|_0^{\infty} + \int_0^{\infty} \frac{1}{2} e^{-\lambda y}\,dy + y \cdot \frac{1}{2} e^{\lambda y}\Big|_{-\infty}^{0} - \int_{-\infty}^{0} \frac{1}{2} e^{\lambda y}\,dy$

$= 0 + \frac{1}{2}(0-1) - 0 + (1-0)\frac{1}{2} + 0$

$= (0-1)\frac{1}{2} - 0 + (1-0)\frac{1}{2} + 0$

$= 0$ ★

(d) $E[Y^2] = \int_0^{\infty} y^2 \cdot \frac{\lambda}{2} e^{-\lambda y}\,dy + \int_{-\infty}^{0} y^2 \cdot \frac{\lambda}{2} e^{\lambda y}\,dy$

$= -y^2 \cdot \frac{1}{2} e^{-\lambda y}\Big|_0^{\infty} + \int_0^{\infty} 2y \cdot \frac{1}{2} e^{-\lambda y}\,dy + y^2 \cdot \frac{1}{2} e^{\lambda y}\Big|_{-\infty}^{0} - \int_{-\infty}^{0} 2y \cdot \frac{1}{2} e^{\lambda y}\,dy$

$= 0 + \int_0^{\infty} y \cdot \lambda e^{-\lambda y}\,dy + 0 - \int_{-\infty}^{0} y \cdot \lambda e^{\lambda y}\,dy$

$= \int_0^{\infty} \frac{\lambda}{\lambda} y \cdot \lambda e^{-\lambda y}\,dy + \int_{-\infty}^{0} \frac{\lambda}{\lambda} y \cdot \lambda e^{\lambda y}\,dy$

$= \int_0^{\infty} \frac{1}{\lambda} \cdot \lambda y \cdot \lambda e^{-\lambda y}\,dy + \int_{-\infty}^{0} \frac{1}{\lambda} \cdot \lambda y \cdot \lambda e^{\lambda y}\,dy$

$= \frac{1}{\lambda} E[Y]$

$\rightarrow \mathrm{Var}[Y] = E[Y^2] - (E[Y])^2$

$= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$ ★

(e) $P(|Y|>a)$

$= 1 - P(Y=a) = 1$ ★

(f) $E[e^{tY}] = \int_0^{\infty} e^{ty} \cdot \frac{\lambda}{2} e^{-\lambda y}\,dy + \int_{-\infty}^{0} e^{ty} \cdot \frac{\lambda}{2} e^{\lambda y}\,dy$

$= \frac{\lambda}{2} \int_0^{\infty} e^{-(\lambda - t)y}\,dy + \frac{\lambda}{2} \int_{-\infty}^{0} e^{(\lambda + t)y}\,dy$

$= \frac{\lambda}{2} \cdot \frac{1}{\lambda - t} e^{-(\lambda - t)y}\Big|_0^{\infty} + \frac{\lambda}{2} \cdot \frac{1}{\lambda + t} e^{(\lambda + t)y}\Big|_{-\infty}^{0}$

$= \frac{\lambda}{2(\lambda - t)}(0 - 1) + \frac{\lambda}{2(\lambda + t)}(1 - 0)$

$= \frac{\lambda}{2}\left(\frac{1}{\lambda + t} - \frac{1}{\lambda - t}\right)$

$= \frac{\lambda}{2} \cdot \frac{-2t}{\lambda^2 - t^2}$

$= \frac{t}{t^2 - \lambda^2}$

$t < \lambda$ 일 때 존재

Prof. Bang Chul Jung

Probability & Random Variables - Midterm Exam

Apr. 22, 2026

Student ID Number: 20220953
Name: 서현준

8.

1. (8 points) Indicate whether each statement is true or false.
   a. If $P(A) = 2P(A^c)$, then $P(A) = 1/2$.
   b. For all A and B, $P(A \cap B) = P(A) \cdot P(B)$.
   c. If $P(A) < P(B)$, then $P(A \cap B) < P(B)$.
   d. If $P(A \cap B) = P(A)$, then $P(A) \geq P(B)$.

(a) $P(A \cup A^c) = P(\Omega) = P(A) + P(A^c) = 3P(A^c) = 1 \rightarrow P(A^c) = \frac{1}{3} \rightarrow P(A) = \frac{2}{3}$ false

(b) $P(A \cap B) = P(A)P(B|A) \rightarrow P(B|A) \neq P(B)$ 인 경우가 있을 수 있으므로 false

(c) $P(A) = P((A \cap B) \cup (A \cap B^c)) = P(A \cap B) + P(A \cap B^c) < P(B) \rightarrow P(A \cap B) < P(B) - P(A \cap B^c) < P(B)$ true

(d) $P(A) = P((A \cap B) \cup (A \cap B^c)) = P(A \cap B) + P(A \cap B^c) \rightarrow P(A) = P(A \cap B)$ 이때 $P(A \cap B^c) = 0$

$P(B) = P((B \cap A) \cup (B \cap A^c)) = P(B \cap A) + P(B \cap A^c)$

$= P(A) + P(B \cap A^c) \geq P(A)$ false

6.

2. (6 points) A person has undertaken a mining job. The probabilities of job completion on time with and without rain are 0.42 and 0.90, respectively. If the probability that it will rain is 0.45, then determine the probability that the mining job will be completed on time.

C : Completion R : rain R' : no rain

$P(C|R) = 0.42$

$P(C|R') = 0.9$

$P(R) = 0.45$

$\rightarrow P(R') = 0.55$

$P(C) = P(C \cap R) + P(C \cap R')$

$= P(R)P(C|R) + P(R')P(C|R')$

$= (0.45) \times (0.42) + (0.55) \times (0.9)$

$= 0.684$

3. (6 points) Let X be a discrete random variable with the following PMF.

6.

$$p_X(x) = \begin{cases} 0.1 & \text{for } x = 0.2 \\ 0.2 & \text{for } x = 0.4 \\ 0.2 & \text{for } x = 0.5 \\ 0.3 & \text{for } x = 0.8 \\ 0.2 & \text{for } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

a. Find $P(X \leq 0.5)$.

b. Find $P(0.25 < X < 0.75)$.

c. Find $P(X = 0.2 \mid X < 0.6)$.

(a) $P(X \leq 0.5) = P_X(0.2) + P_X(0.4) + P_X(0.5) = 0.1 + 0.2 + 0.2 = 0.5$

(b) $P(0.25 < X < 0.75) = P_X(0.4) + P_X(0.5) = 0.2 + 0.2 = 0.4$

(c) $P(X = 0.2 \mid X < 0.6) = \dfrac{P(X = 0.2)}{P(X < 0.6)} = \dfrac{0.1}{0.1 + 0.2 + 0.2} = \dfrac{0.1}{0.5} = 0.2$ ☆

4. (16 points) Random variables X and Y have the joint PMF in the following table.

16

$P_{X,Y}(x,y)$	$y = -1$	$y = 0$	$y = 1$	
$x = -1$	3/16	1/16	0	→ 4/16
$x = 0$	1/6	1/6	1/6	→ 1/2
$x = 1$	0	1/8	1/8	→ 1/4
	17/48	14/48		

a. Are X and Y independent? (Discuss it with proof.)

$\frac{3+8+6}{48} = \frac{17}{48}$ 34+

$P_{X,Y}(-1,-1) = \frac{3}{16}$

$P_X(-1) = \frac{4}{16}$

$P_Y(-1) = \frac{17}{48}$

$P_{X,Y}(-1,-1) \neq P_X(-1)\,P_Y(-1)$ 이므로 X ☆